Roman Huts

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Summary

Senior Software Engineer specializing in C++, GPU acceleration, video processing, and cross-platform SDK development. Track record of shipping developer-facing technologies across Windows and Linux including AMD AMF SDK, AV1 integration into OBS, and AI-powered video processing. Experienced in bridging hardware and software, owning the SDK release management lifecycle, and delivering technologies adopted by external developers and open-source communities.

Experience

Senior Software Engineer, Developer Solutions 2022 – Present
AMD *Markham, ON / Remote*

- Delivered **14** public AMF SDK releases, resolved **80+ external developer issues**, and hundreds of internal bugs.
- Improved encoding efficiency by 30% for AMD GPU users by integrating AMD's **AV1 encoder** into OBS with around **20 million users (C/C++).**
- Developed **Flatpak** runtime packaging to bring AMD AMF support to the **freedesktop.org runtime** ecosystem, enabling AMD GPU-accelerated video encoding for Linux applications and reaching **679K+ monthly downloads** on Flathub.
- Designed and implemented AI **model compression and DLL embedding** to enable efficient distribution of an AMD AI video upscaling solution, and refactored the component architecture to support future extensibility.
- Identified, designed, and delivered **video rotation** support in the AMF VideoConverter, addressing a capability gap and strengthening the SDK's competitiveness in video processing capabilities.

Software Development Engineer 2, Developer Solutions 2020 – 2022
AMD *Markham, ON / Remote*

- Improved Pre-Analysis and Pre-Processing component performance by up to **15**× by implementing **hardware-accelerated** versions of CPU-based algorithms **(OpenCL / DX11 / DX12 / Vulkan / C++)** in the AMF SDK, improving the user experience for up to **285 million users** of AMD GPU drivers worldwide.
- Developed ONNX model-based audio noise reduction demos to showcase AMD technology to Lenovo as part of a product sales initiative, extended support to Linux and integrated reusable components into the AMF SDK.

Software Developer Intern, Game Engine Integration 2018 – 2019
AMD *Markham, ON*

- Improved SDK robustness by fixing over 20 bugs for AMF and LiquidVR SDKs such as eGPU support **(C++).**
- Expanded support for AMD products by integrating **FreeSync 2 HDR** into **Unreal Engine 4 (C++)**.
- Owned the team's Jenkins CI/CD farm, maintaining **98%** uptime and improving automation reliability **(Python / Bash / Groovy)**.

Business Projects

NDIO Customs | *Node, Javascript, electronics, lithium battery systems*

- Founded and developed a custom electric motorcycle business, combining software development, electronics integration, and lithium-ion battery system design. Earned **1**st **place** in the Special Construction class at the North American International Motorcycle Supershow(2023 & 2025).

Music Licensing Lead Generation (DCSIL Program) | *Android Studio, Java, Flask, Python, Heroku, Redis, MongoDB*

- Led a team to develop a cross-platform application for non-confrontational lead generation, driving development from concept to MVP technical demo.
- Presented to music licensing industry stakeholders, including **BMAT USA** and **Dataclef** (a **SOCAN** subsidiary), to demonstrate its potential and gather industry feedback.

Awards

- CG Recognition award for shipping **AV1** with **OBS** (AMD 2022)
- Spotlight award for role in **perforce to git migration** (AMD 2021)
- Recognition award for **outstanding** co-op **mentorship** (AMD 2021)

Technical Skills

Languages: C++, C, Python, Bash, JavaScript, SQL, HTML, CSS
Computer Graphics: OpenCL, DirectX11, Vulkan, DirectX12
Developer Tools: CMake, Make, Visual Studio, Git, GitHub, Perforce, JIRA, Confluence, GDB, GPUView, Valgrind, RadeonGPU Analyzer, Elecard StreamEye Stream Analyzer, ONNX Runtime, Ultralytics, OpenCV, CodeQL,
Operating Systems: Linux, Windows

Education

University of Toronto Toronto, ON
Honors Bachelor of Science with Distinction, Computer Science Specialist